

Question 1 [40pts]: Bayesian Networks and Variable Elimination

Q1a [10 pts]

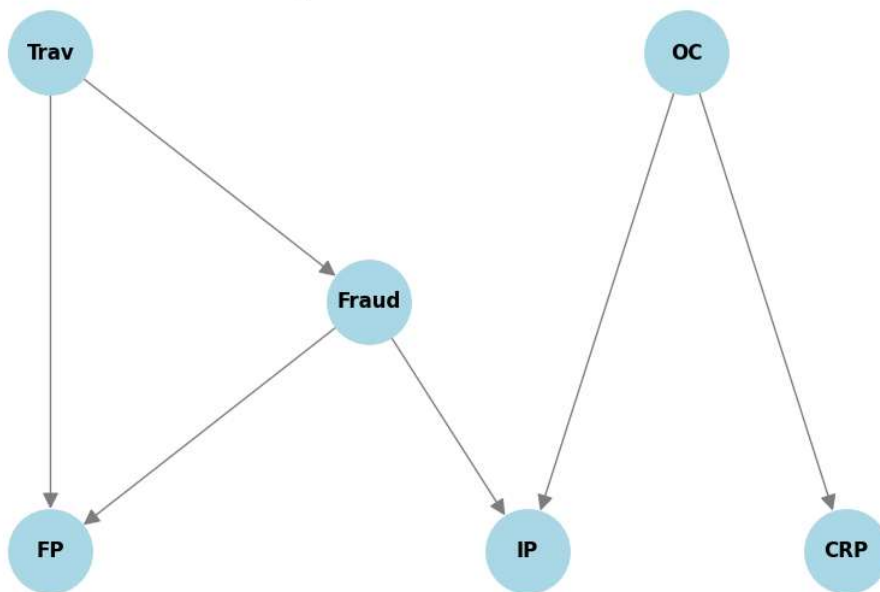
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In [1]: import networkx as nx
import matplotlib.pyplot as plt

G = nx.DiGraph()
nodes = ['Trav', 'OC', 'Fraud', 'FP', 'IP', 'CRP']
edges = [('Trav', 'Fraud'), ('Trav', 'FP'), ('Fraud', 'FP'),
        ('OC', 'IP'), ('Fraud', 'IP'), ('OC', 'CRP')]
G.add_nodes_from(nodes)
G.add_edges_from(edges)

pos = {
    'Trav': (0, 2),
    'OC': (4, 2),
    'Fraud': (2, 1),
    'FP': (0, 0),
    'IP': (3, 0),
    'CRP': (5, 0)
}

plt.figure(figsize=(8, 5))
nx.draw(G, pos, with_labels=True, node_color='lightblue',
        node_size=2500, font_size=12, font_weight='bold',
        arrows=True, arrowsize=20, edge_color='gray')
plt.title('Bayes Network for Fraud Detection')
plt.show()
```

Bayes Network for Fraud Detection



Root nodes (no parents):

Trav	P(Trav)
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T	0.05
---	------

F	0.95
---	------

OC	P(OC)
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T	0.60
---	------

F	0.40
---	------

Fraud — parents: Trav

Trav	P(Fraud=T)	P(Fraud=F)
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T	0.01	0.99
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F	0.004	0.996
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FP — parents: Trav, Fraud

Trav	Fraud	P(FP=T)	P(FP=F)
T	T	0.90	0.10
T	F	0.90	0.10
F	T	0.10	0.90
F	F	0.01	0.99

IP — parents: OC, Fraud

OC	Fraud	P(IP=T)	P(IP=F)
T	T	0.02	0.98
T	F	0.01	0.99
F	T	0.011	0.989
F	F	0.001	0.999

CRP — parent: OC

OC	P(CRP=T)	P(CRP=F)
T	0.10	0.90
F	0.001	0.999

Q1b [25 pts]

Queries:

- $P(\text{Fraud})$
- $P(\text{Fraud} \mid FP, \neg IP, CRP)$

Initial factors (from Q1a CPTs):

- $f_1(\text{Trav}) = P(\text{Trav})$
- $f_2(\text{OC}) = P(\text{OC})$
- $f_3(\text{Fraud}, \text{Trav}) = P(\text{Fraud} \mid \text{Trav})$
- $f_4(\text{FP}, \text{Trav}, \text{Fraud}) = P(\text{FP} \mid \text{Trav}, \text{Fraud})$
- $f_5(\text{IP}, \text{OC}, \text{Fraud}) = P(\text{IP} \mid \text{OC}, \text{Fraud})$
- $f_6(\text{CRP}, \text{OC}) = P(\text{CRP} \mid \text{OC})$

Elimination order: Trav, FP, ~~Fraud~~ (query), IP, OC, CRP

Query 1: $P(\text{Fraud})$

Marginalize the joint via BN factorization:

$$P(\text{Fraud}) = \sum_{\text{Trav}} \sum_{\text{FP}} \sum_{\text{IP}} \sum_{\text{OC}} \sum_{\text{CRP}} P(\text{Trav}) P(\text{OC}) P(\text{Fraud} \mid \text{Trav}) P(\text{FP} \mid \text{Trav}, \text{Fraud}) P(\text{IP} \mid \text{OC}, \text{Fraud}) P(\text{CRP} \mid \text{OC})$$

Each CPT term is a factor (f_1 through f_6). VE eliminates one variable at a time.

Step 1 — eliminate Trav. Factors containing Trav: f_1, f_3, f_4 .

$$f_7(\text{Fraud}, \text{FP}) = \sum_{\text{Trav}} f_1(\text{Trav}) \cdot f_3(\text{Fraud}, \text{Trav}) \cdot f_4(\text{FP}, \text{Trav}, \text{Fraud})$$

Fraud	FP	f_7
T	T	0.000830
T	F	0.003470
F	T	0.054012
F	F	0.941688

Step 2 — eliminate FP. Factor: f_7 .

$$f_8(\text{Fraud}) = \sum_{\text{FP}} f_7(\text{Fraud}, \text{FP})$$

Fraud	f_8
T	0.004300
F	0.995700

Step 3 — eliminate IP. Factor: f_5 .

$$f_9(\text{OC}, \text{Fraud}) = \sum_{\text{IP}} f_5(\text{IP}, \text{OC}, \text{Fraud}) = 1 \ \forall \ (\text{OC}, \text{Fraud})$$

CPT sums to 1 over IP — no evidence restricts it.

Step 4 — eliminate OC. Factors: f_2, f_6 . ($f_9 = 1$, drops out.)

$$f_{10}(\text{CRP}) = \sum_{\text{OC}} f_2(\text{OC}) \cdot f_6(\text{CRP}, \text{OC})$$

CRP	f_{10}
T	0.0604
F	0.9396

Step 5 — eliminate CRP. Factor: f_{10} .

$$f_{11} = \sum_{\text{CRP}} f_{10}(\text{CRP}) = 0.0604 + 0.9396 = 1.0$$

Result: $P(\text{Fraud}) = f_8(\text{Fraud}) \cdot f_{11} = f_8(\text{Fraud})$

$P(\text{Fraud} = T) = 0.0043$	$P(\text{Fraud} = F) = 0.9957$
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Query 2: $P(\text{Fraud} \mid FP = T, IP = F, CRP = T)$

Conditional probability definition, then BN factorization:

$$P(\text{Fraud} \mid FP, \neg IP, CRP) = \frac{P(\text{Fraud}, FP, \neg IP, CRP)}{P(FP, \neg IP, CRP)} \propto P(\text{Fraud}, FP, \neg IP, CRP)$$

Denominator doesn't depend on Fraud — normalizing constant α .

Expand numerator — marginalize Trav and OC:

$$P(\text{Fraud}, FP, \neg IP, CRP) = \sum_{\text{Trav}} \sum_{\text{OC}} P(\text{Trav}) P(\text{OC}) P(\text{Fraud} \mid \text{Trav}) P(FP=T \mid \text{Trav}, \text{Fraud}) P(IP=F \mid \text{OC}, \text{Fraud}) P(CRP=T \mid \text{OC})$$

Restrict factors to evidence, sum out hidden variables.

Evidence restriction:

f_4 restricted to $FP = T \rightarrow f'_4(\text{Trav}, \text{Fraud})$:

Trav	Fraud	f'_4
T	T	0.90
T	F	0.90
F	T	0.10
F	F	0.01

f_5 restricted to $IP = F \rightarrow f'_5(\text{OC}, \text{Fraud})$:

OC	Fraud	f'_5
T	T	0.98
T	F	0.99
F	T	0.989
F	F	0.999

f_6 restricted to $CRP = T \rightarrow f'_6(\text{OC})$:

OC	f'_6
T	0.10
F	0.001

Eliminate hidden variables: Trav, OC.

Step 1 — eliminate Trav. Factors: f_1, f_3, f'_4 .

$$f_7(\text{Fraud}) = \sum_{\text{Trav}} f_1(\text{Trav}) \cdot f_3(\text{Fraud}, \text{Trav}) \cdot f'_4(\text{Trav}, \text{Fraud})$$

Fraud	computation	f_7
T	0.05×0.01	0.000830
	$\times 0.90$	
	$+ 0.95$	
	$\times 0.004$	
	$\times 0.10$	
F	0.05×0.99	0.054012
	$\times 0.90$	
	$+ 0.95$	
	$\times 0.996$	
	$\times 0.01$	

Step 2 — eliminate OC. Factors: f_2, f'_5, f'_6 .

$$f_8(\text{Fraud}) = \sum_{\text{OC}} f_2(\text{OC}) \cdot f'_5(\text{OC}, \text{Fraud}) \cdot f'_6(\text{OC})$$

Fraud	computation	f_8
T		0.059196

Fraud	computation	f_8
	0.60×0.98 $\times 0.10$ $+ 0.40$ $\times 0.989$ $\times 0.001$	
F	0.60×0.99 $\times 0.10$ $+ 0.40$ $\times 0.999$ $\times 0.001$	0.059800

Multiply and normalize:

$$P(\text{Fraud} \mid e) \propto f_7(\text{Fraud}) \cdot f_8(\text{Fraud})$$

Fraud	$f_7 \times f_8$	P (Fraud evidence)
T	$0.000830 \times 0.059196 = 0.0000491$	0.0150
F	$0.054012 \times 0.059800 = 0.003230$	0.9850

$$\alpha = 1/(0.0000491 + 0.003230) = 1/0.003279$$

$P(\text{Fraud} = T \mid FP, \neg IP, CRP) \approx 0.0150$

Prior 0.43% → posterior 1.50%, ~3.5× — foreign purchase is the main driver.

Q1c [5 pts]

Action: Before making the internet purchase, buy a computer-related item with the stolen card (set $CRP = T$).

Why this works: $CRP = T \rightarrow P(OC = T)$ goes up → for computer owners, internet purchases are more normal (1% vs 0.1%) → $IP = T$ less suspicious → $P(\text{Fraud})$ drops.

Compare $P(\text{Fraud} \mid IP)$ (without action) vs $P(\text{Fraud} \mid IP, CRP)$ (with action).

Steps 1–2 (eliminate Trav, FP) unchanged from Q1b Query 1: $f_8(\text{Fraud} = T) = 0.0043$, $f_8(\text{Fraud} = F) = 0.9957$.

Restrict f_5 to $IP = T \rightarrow f'_5(OC, \text{Fraud})$:

OC	Fraud	f'_5
T	T	0.02
T	F	0.01
F	T	0.011
F	F	0.001

Eliminate OC (with unrestricted f_6): $f_9(\text{Fraud}, CRP) = \sum_{OC} f_2(OC) \cdot f'_5(OC, \text{Fraud}) \cdot f_6(CRP, OC)$

Fraud	CRP	computation	f_9
T	T	0.60×0.02 $\times 0.10$ $+ 0.40$ $\times 0.011$ $\times 0.001$	0.001204
T	F	0.60×0.02 $\times 0.90$ $+ 0.40$ $\times 0.011$ $\times 0.999$	0.015196
F	T	0.60×0.01 $\times 0.10$ $+ 0.40$ $\times 0.001$ $\times 0.001$	0.000600
F	F	0.60×0.01 $\times 0.90$ $+ 0.40$ $\times 0.001$ $\times 0.999$	0.005800

Without action — $P(\text{Fraud} \mid IP)$: sum out CRP from f_9 :

$$f_{10}(\text{Fraud} = T) = 0.001204 + 0.015196 = 0.016400$$

$$f_{10}(\text{Fraud} = F) = 0.000600 + 0.005800 = 0.006400$$

$$P(\text{Fraud} \mid IP) \propto f_8 \times f_{10}:$$

Fraud	$f_8 \times f_{10}$	P (Fraud IP)
T	$0.0043 \times 0.0164 = 0.0000705$	0.0109

Fraud	$f_8 \times f_{10}$	P (Fraud IP)
F	$0.9957 \times 0.0064 = 0.006372$	0.9891

With action — $P(\text{Fraud} \mid IP, CRP)$: take $CRP = T$ row of f_9 directly (no CRP summation needed):

$$f_9(\text{Fraud} = T, CRP = T) = 0.001204$$

$$f_9(\text{Fraud} = F, CRP = T) = 0.000600$$

$$P(\text{Fraud} \mid IP, CRP) \propto f_8 \times f_9(\cdot, T):$$

Fraud	$f_8 \times f_9$	P (Fraud IP , CRP)
T	$0.0043 \times 0.001204 = 0.00000518$	0.0086
F	$0.9957 \times 0.000600 = 0.000598$	0.9914

Result: fraud probability $0.0109 \rightarrow 0.0086$ — decrease of 0.0023 ($\approx 21\%$ reduction).

$$P(\text{Fraud} \mid IP) \approx 0.0109 \xrightarrow{\text{buy CRP}} P(\text{Fraud} \mid IP, CRP) \approx 0.0086$$

CS 486 A2 — Q2 Q3

2. Markov Decision Processes (60 points)

Q2a [20 pts]

$R(s) = -1.0$ for all non-terminal states. $\gamma = 1$. Value iteration until convergence (3 decimal places).

Utility values per iteration:

Iteration 0:

0.000	0.000	0.000	0.000
0.000	X	0.000	-1.000
0.000	0.000	0.000	1.000

Iteration 1:

-1.000	-1.000	-1.000	-1.000
-1.000	X	-1.000	-1.000
-1.000	-1.000	-0.200	1.000

Iteration 2:

-2.000	-2.000	-2.000	-2.000
-2.000	X	-1.360	-1.000
-2.000	-1.360	-0.320	1.000

Iteration 3:

-3.000	-3.000	-2.488	-2.200
-3.000	X	-1.492	-1.000
-2.488	-1.528	-0.368	1.000

Iteration 4:

-4.000	-3.590	-2.714	-2.269
-3.590	X	-1.544	-1.000
-2.771	-1.600	-0.386	1.000

Iteration 5:

-4.631	-3.889	-2.821	-2.298
-3.935	X	-1.563	-1.000
-2.916	-1.629	-0.393	1.000

Iteration 6:

-4.968	-4.034	-2.869	-2.312
-4.120	X	-1.571	-1.000
-2.988	-1.640	-0.396	1.000

Iteration 7:

-5.136	-4.102	-2.891	-2.318
-4.215	X	-1.574	-1.000
-3.023	-1.645	-0.397	1.000

Iteration 8:

-5.217	-4.133	-2.901	-2.321
-4.261	X	-1.575	-1.000
-3.039	-1.646	-0.397	1.000

Iteration 9:

-5.255	-4.147	-2.905	-2.322
-4.284	X	-1.575	-1.000
-3.047	-1.647	-0.397	1.000

Iteration 10:

-5.272	-4.154	-2.907	-2.323
-4.294	X	-1.575	-1.000
-3.051	-1.647	-0.397	1.000

Iteration 11:

-5.279	-4.156	-2.908	-2.323
-4.299	X	-1.575	-1.000
-3.052	-1.647	-0.397	1.000

Iteration 12:

-5.283	-4.158	-2.908	-2.323
-4.302	X	-1.575	-1.000
-3.053	-1.647	-0.397	1.000

Iteration 13:

-5.284	-4.158	-2.908	-2.323
-4.303	X	-1.575	-1.000
-3.053	-1.647	-0.397	1.000

Iteration 14:

-5.285	-4.158	-2.908	-2.323
-4.303	X	-1.575	-1.000
-3.053	-1.647	-0.397	1.000

Iteration 15 (converged - same as iteration 14):

-5.285	-4.158	-2.908	-2.323
-4.303	X	-1.575	-1.000
-3.053	-1.647	-0.397	1.000

Optimal policy (R(s) = -1.0):

	col1	col2	col3	col4
row1	→	→	↓	↓
row2	↓	X	↓	-1
row3	→	→	→	+1

Q2b [20 pts]

Three policy ranges found. Rows 2-3 stay the same across all R values — only row 1 changes.

Range 1: $R(s) = -1.6$

	col1	col2	col3	col4
row1	→	→	→	↓
row2	↓	X	↓	-1
row3	→	→	→	+1

$s(1,3) = \rightarrow$ — agent at (1,3) heads right toward $s(1,4)$ then down to the -1 terminal. Living cost is so high (-1.6/step) that reaching any terminal fast beats a longer path to +1.

Transition: $R = -1.6 \rightarrow R = -1.5$ $s(1,3)$ changes from $\rightarrow$ to $\downarrow$. Agent now goes down toward +1 instead of right toward -1.

Range 2: $R(s) = -1.5$ to -0.8

	col1	col2	col3	col4
row1	→	→	↓	↓
row2	↓	X	↓	-1
row3	→	→	→	+1

$s(1,1) = \rightarrow$ — still heading right, taking a slightly longer path toward +1 via row 3.

Transition: $R = -0.8 \rightarrow R = -0.7$ $s(1,1)$ changes from $\rightarrow$ to $\downarrow$. Agent now goes directly down — shorter path to row 3 and the +1 terminal.

Range 3: $R(s) = -0.7$ to -0.5

	col1	col2	col3	col4
row1	↓	→	↓	↓
row2	↓	X	↓	-1
row3	→	→	→	+1

All non-terminal states now route toward the +1 terminal. Lower living penalty gives the agent more room to avoid -1.

Q2c [20 pts]

Two policy ranges found. Again only row 1 changes — rows 2-3 identical across all R values.

Range 1: $R(s) = -0.08$ to -0.05

	col1	col2	col3	col4
row1	↓	←	↓	←
row2	↓	X	↓	-1
row3	→	→	→	+1

$s(1,3) = \downarrow$ — goes down toward row 3 and the +1 terminal. $s(1,4) = \leftarrow$ — actively moves away from the -1 terminal. Living penalty is small enough that the agent takes longer paths to avoid -1.

Transition: $R = -0.05 \rightarrow R = -0.04$ $s(1,3)$ changes from $\downarrow$ to $\leftarrow$. Agent shifts from going down to going left — even more avoidance of the -1 side of the grid.

Range 2: $R(s) = -0.04$ to -0.03

	col1	col2	col3	col4
row1	↓	←	←	←
row2	↓	X	↓	-1
row3	→	→	→	+1

$s(1,3)$ now also $\leftarrow$ — nearly zero living cost, so the agent strongly prefers the longest safe route to +1 over any path near -1.

3. Bonus: Viterbi Decoding (20 extra points)

Solved by program — see code.py.

HMM parameters: $P(s_0)=0.4$, $P(st|st-1)=0.7$, $P(st|\text{not } st-1)=0.2$, $P(ot|st)=0.9$, $P(ot|\text{not } st)=0.2$. Observations: $O_0=T$, $O_1=T$, $O_2=T$, $O_3=F$.

Step-by-step delta values and back-pointers:

Step	S=True delta	S=True backptr	S=False delta	S=False backptr
0	0.36	-	0.12	-

Step	S=True delta	S=True backptr	S=False delta	S=False backptr
1	0.2268	True	0.0216	True
2	0.142884	True	0.013608	True
3	0.01000188	True	0.03429216	True

Termination: $\max(0.01000188, 0.03429216) = 0.03429216$ at S3=False.

Traceback: S3=F, backptr=T -> S2=T, backptr=T -> S1=T, backptr=T -> S0=T.

Optimal hidden state sequence: (T, T, T, F)